Kaustuv Dey

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Education

Dr. B R Ambedkar National Institute of Technology — *Aug. 2024 – June 2026*
M.Tech in IT (Data Analytics) | Current SGPA: 8.00 | CGPA: 7.80 — Jalandhar, Punjab

ICFAI University, Tripura — *Sept. 2020 – June 2024*
B.Tech in Computer Science & Engineering | CGPA: 8.76 — Agartala, Tripura

Experience

Warner Bros. Discovery — *Sept. 2025 - Present*
Software Engineer Intern — Bangalore, India

- Engineered architectural optimizations on **caption & script** generation pipelines for **WBD movies and episodics**, parallelizing compute stages — cutting runtime by **75%** and saving **$8,000+/yr** in cloud cost.
- Instrumented runtime task monitoring across content workflows which dynamically adjusts per-job limits based on asset duration — reducing required on-call effort, false-negative alerts by **75%**, and customer SLA breaches.
- Designed a new **localization-demand payload** across the **translation** pipeline, aligning cross-functional team services (ingest, orchestrator, delivery) to ship a subtitle translation product with parallel **multi-artifact support**.
- Provisioned scalable **VM environments (GCP Workbenches)** and orchestrated containerized inference workloads (OCR, LLM-based dialog annotation) via **Vertex AI** Custom Jobs, improving compute efficiency.
- Improved **production-deployed** pipeline reliability and deployment velocity by shipping **7+** production features and **80+** unit tests under **CI/CD**, and integrating PagerDuty & Slack alerting — reducing on-call resolution times.
- Diagnosed and hot-fixed **20+** cross-service production incidents spanning **5+ microservices**, restoring pipeline availability and preventing downstream content-delivery delays.

GAO Tek Inc. — *Aug. 2022 - Apr. 2023*
Software Engineering Intern — Team Lead — Ontario, Canada

- Optimized C code for 3 RFID tracking systems, improving accuracy by **85%** and reducing errors by **40%**.
- Automated the reporting workflow for a **40+** engineer team using Excel macros and Bitrix24 integrations, cutting reporting time by **60%**.
- Trained and mentored **6** junior developers in RFID and C programming, achieving 100% certification in 3 months.

ZIROH Labs — *Jan. 2022 - Mar. 2022*
Software Development Intern — Bangalore, India

- Developed **AES** encryption for **JSON** data in a **Maven-based** project, ensuring robust security. Integrated **REST APIs** for secure communication and **SQLite** for efficient encrypted data storage, improving performance by **20%**.

Projects

Captioning Localization Quality Suite | *Warner Bros. Discovery*

- Developed a video-understanding pipeline to extract and classify on-screen text via **scene text recognition with LLM post-processing**, benchmarked on **70+** video-script pairs with **5,300+** annotated text blocks.
- Engineered a **burnt-in-subtitle filtering** module that cleans transcription output of visual-overlay noise while preserving genuine dialogue, unblocking reliable speech-to-text quality benchmarking for the team.

VisionTrack: AI-Powered CCTV Search & Geotagging System | *Hackathon Project*

- Built an intelligent CCTV search and geotagging system leveraging a **Siamese model** for image similarity and **Google Maps API** for location mapping interacted using a simple website build on **Django** for investigators.
- Integrated face, location, number plate recognition (using **YOLO** model) in both image and video format, to support Tripura State Police in criminal investigations and crime pattern analysis.

My Kernel |

- Developed a simple Unix-like Kernel in **C** and **Assembly** simulated on **QEMU**, featuring system calls, paging.
- Built low-level **OS** components including a **Multiboot-compliant** bootloader, interrupt handling with a **Docker** build environment for portability.

Technical Skills

- **Programming Languages:** Python, C++, SQL, C, Bash/Shell, Golang, JavaScript
- **Tools/DevOps/Cloud/Monitoring:** Git, Docker, Linux, CI/CD, GCP, JIRA, PagerDuty, QEMU
- **Data/ML:** Airflow, Vertex AI, BigQuery, Pandas, NumPy

Achievements

- **2 research papers published on Quantum Steganography** in Elsevier FTNCT 08 and Springer BIDA 03.
- **GATE 2024 Qualified** in Computer Science and Information Technology with **91.36 percentile**. *2024*
- **1st Runner-up** in Tripura Police Tech hackathon 2022-23 among **150 + teams** all over India. *2022-23*
- **Start-Up IOT Project Funded (Rs. 3 lakhs)** under New Generation Inovation Network (NGIN) by Govt. of Tripura.
- **Specialist** at Codeforces & **5 star** in C at Hackerrank , solved 500+ problems across coding platforms.